

Lecture 10

About Fourier Series Convergence

$$\begin{array}{ccc} x(t) & \longrightarrow & c_n = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt \\ \begin{array}{l} T\text{-periodic} \\ \text{(Riemann integrable)} \end{array} & & \omega = \frac{2\pi}{T} \end{array}$$

Here, the values of $c_n, n \in \mathbb{Z}$, is called the spectral description of $x(t)$. This is also known as Fourier or Frequency-Domain description.

Question: Does the spectral description contain all the information about $x(t)$?

That is, given $c_n, n \in \mathbb{Z}$, is it possible to find $x(t)$ for all t ?

I. Local Convergence Result

Reference: Stein and Shakarchi
"Fourier Analysis: An Introduction"
(Theorem 2.1)

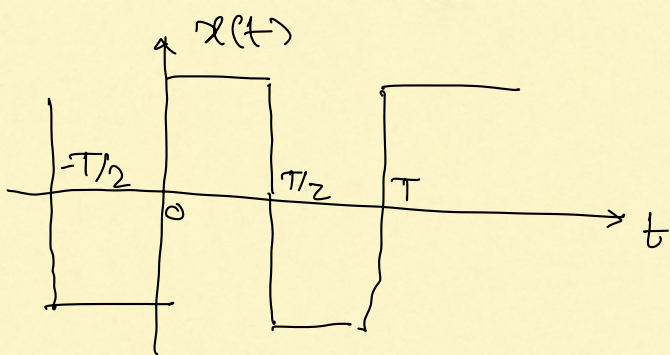
Thm 1: If x is differentiable at a point t_0 , then:

$$S_N(t_0) \rightarrow x(t_0) \text{ as } N \rightarrow \infty$$

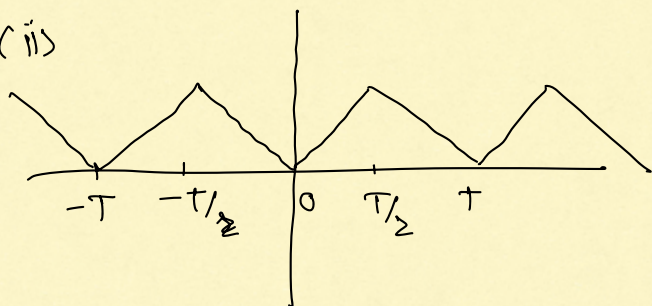
$$\lim_{N \rightarrow \infty} [S_N(t_0)] = x(t_0).$$

$$S_N(t_0) = \sum_{n=-N}^{+N} c_n e^{jn\omega t_0}.$$

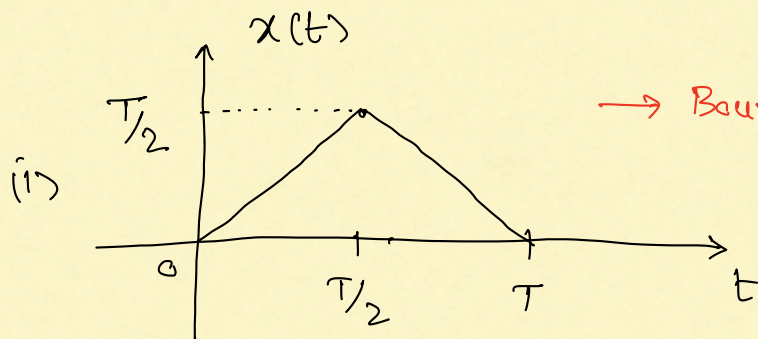
(i)



(ii)

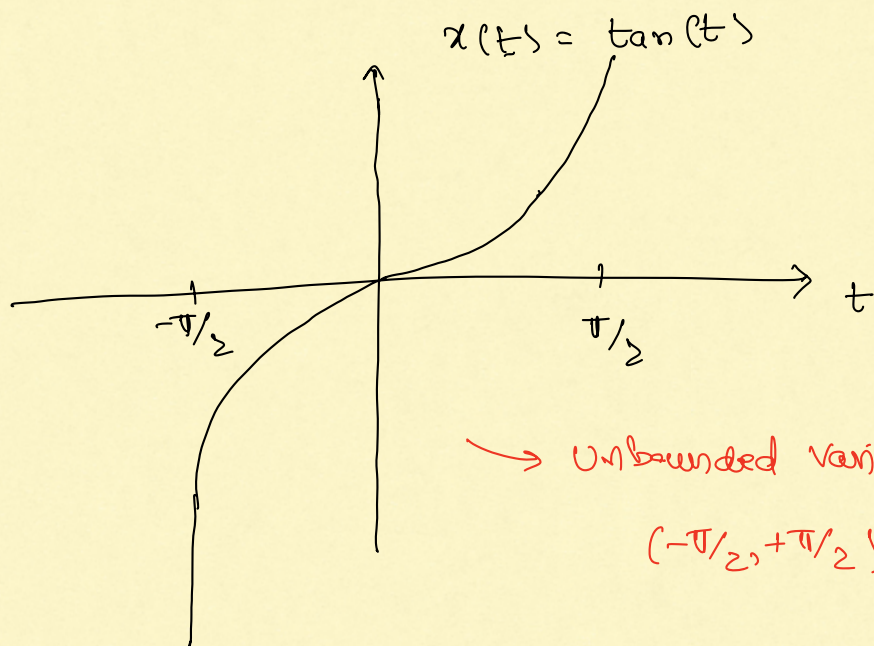


II. Bounded Variation



→ Bounded Variation
 $[0, \pi]$.

(ii)



→ Unbounded Variation
 $(-\pi/2, \pi/2)$

(iii) $x(t) = \sin(2\pi \tan(t))$.

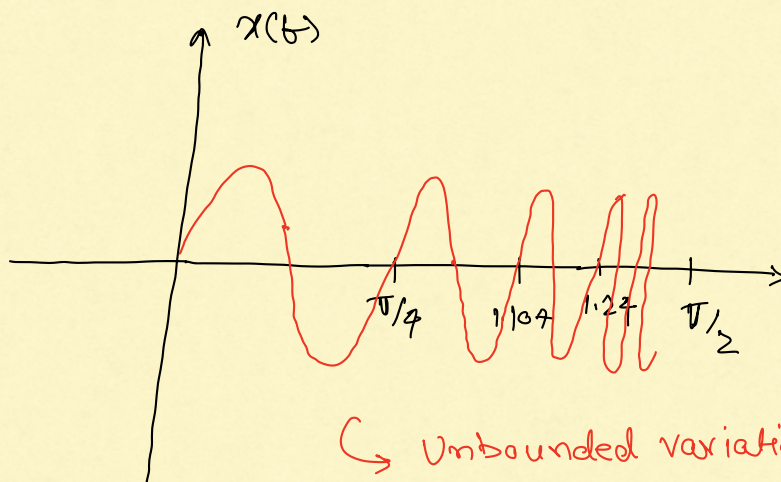
in the interval $0 \leq t < \pi/2$

Recall that: one cycle of $\sin$ happens as its argument changes by 2π .

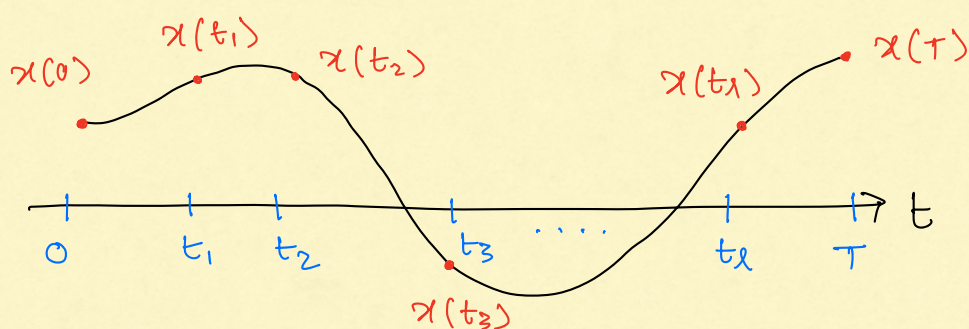
i.e., as $\tan(t)$ increments from l to $l+1$ for $l \in \mathbb{Z}$, $l \geq 0$, we have one cycle of $\sin$.

t	$\tan(t)$
0	0
0.78 $\approx \pi/4$	1
≈ 1.107	2
≈ 1.107	3
$\vdots$	$\vdots$
$\tan^{-1}(l)$	l

$\tan^{-1}(l) \in (0, \pi/2)$
for all $l \geq 1$.



Formal Definition of Bounded Variation (in $[c, T]$)



$$|x(t_1) - x(0)| + |x(t_2) - x(t_1)| + \dots$$

$$\dots + |x(t_l) - x(t_{l-1})| + |x(T) - x(t_l)|$$

Defn: We say that x is of Bounded variation in $[0, T]$ if:

there exists a constant $M > 0$ such that for any $l \geq 1$ and any $0 < t_1 < t_2 < \dots < t_l < T$ we have

$$|x(t_1) - x(0)| + |x(t_2) - x(t_1)| + \dots$$

$$\dots + |x(T) - x(t_l)| \leq M.$$

III. The Dirichlet-Jordan Test

Thm 2: If x is of bounded variation in $[0, T]$ then:

i) If x is continuous at t_0 :

$$\lim_{N \rightarrow \infty} \left[S_N(t_0) \right] = x(t_0)$$

ii) If x has a discontinuity at t_0 :

$$\begin{aligned} \lim_{N \rightarrow \infty} \left[S_N(t_0) \right] &= \text{mid-point of the discontinuity} \\ &= \frac{x(t_0^-) + x(t_0^+)}{2} \end{aligned}$$

Quiz: 9am on 02 Feb

Syllabus: Lec 5-9

Tutorial: Friday 6:30pm

Venue: TBD.